

WIRELESS COMMUNICATION

Types of Wireless Communication



Fig. 1. The Zenith model 1100 console-controlled set shown here is capable of displaying all of the TV set's best picture action made for picture and sound.

By
WALTER H. ECHMEAUM
Television Consultant
RADIO & TELEVISION NEWS

In many cases these units are furnished with the set, but, they can be installed by any service technician.

Remote Control for TV

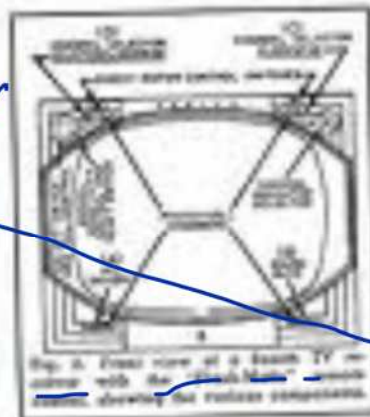
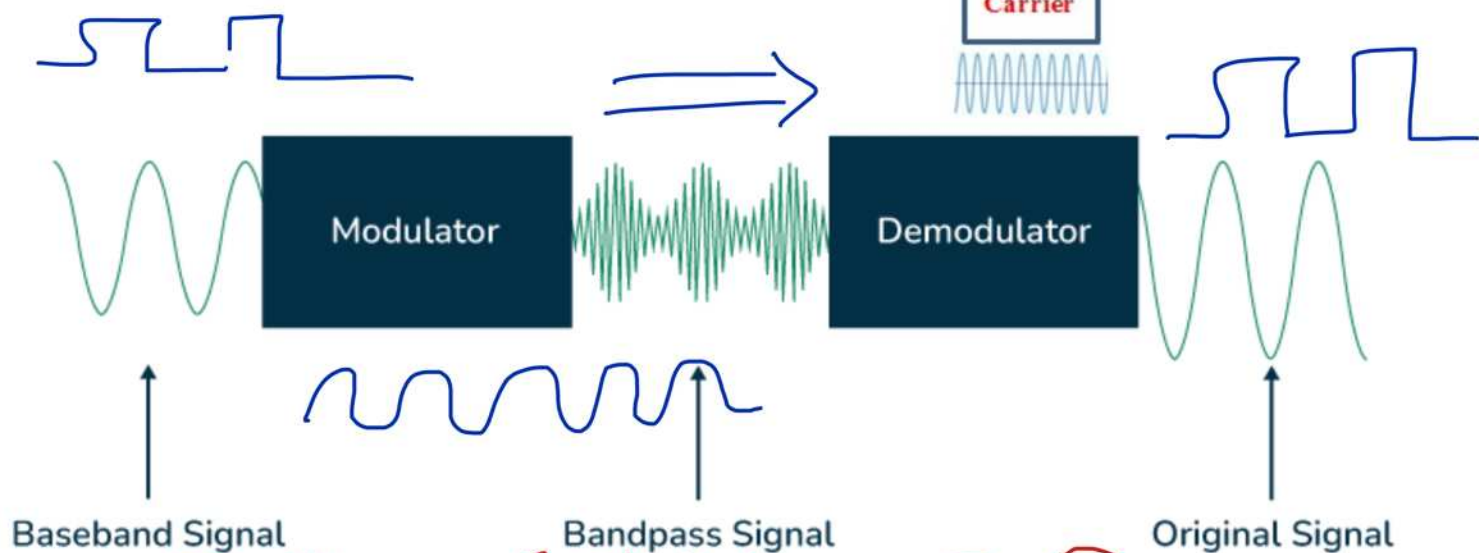
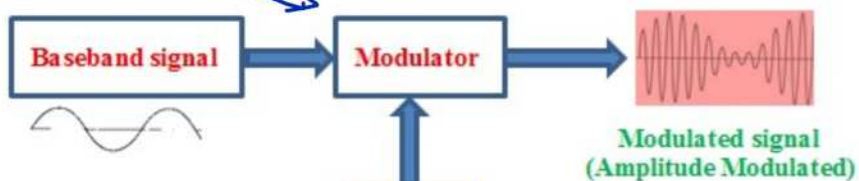
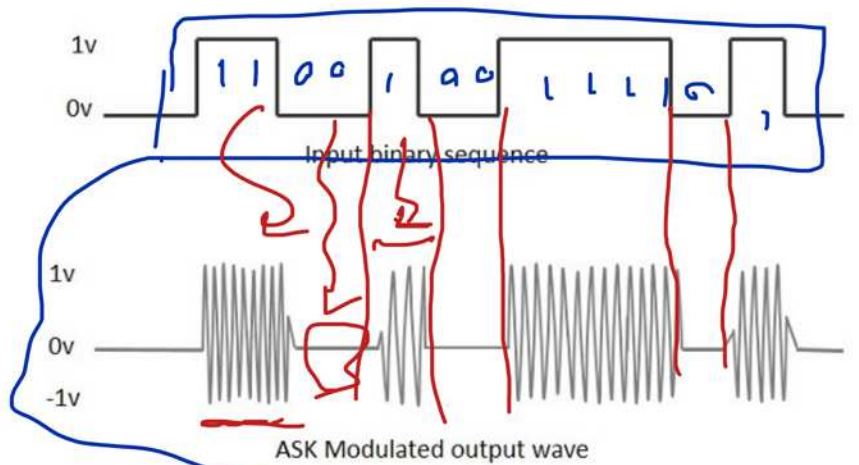
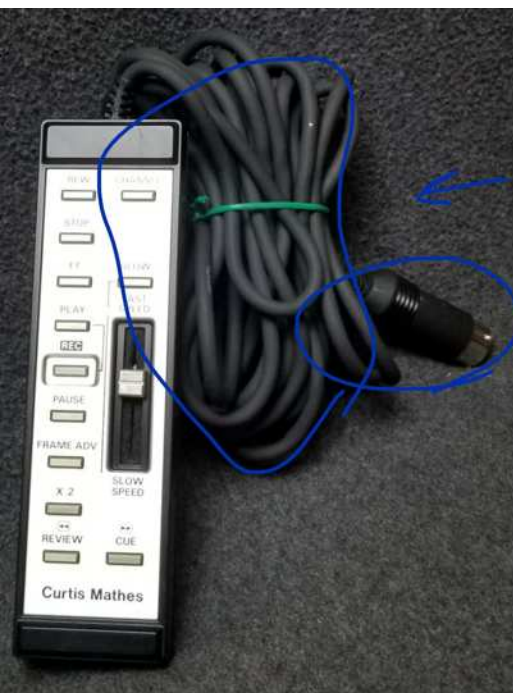


Fig. 2. Front view of a Zenith TV remote with the "Zenith" brand name, showing the remote components.





#	Frequency Range	Antenna Length in mm	Antenna Length in inches
1	169 - 199 MHz	388 mm	15.28"
2	199 - 235 MHz	329 mm	12.95"
3	470 - 548 MHz	140 mm	5.51"
4	548 - 663 MHz	120 mm	4.72"
5	638 - 738 MHz	104 mm	4.09"
6	738 - 865 MHz	89 mm	3.50"
7	902 - 1015 MHz	74 mm	2.92"
8	1045 - 1154 MHz	65 mm	2.55"
9	1240 - 1260 MHz	57 mm	2.24"
10	1350 - 1525 MHz	49 mm	1.93"

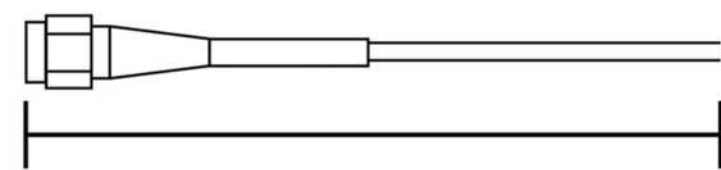
170 MHz → 200 MHz

90 cm

115200

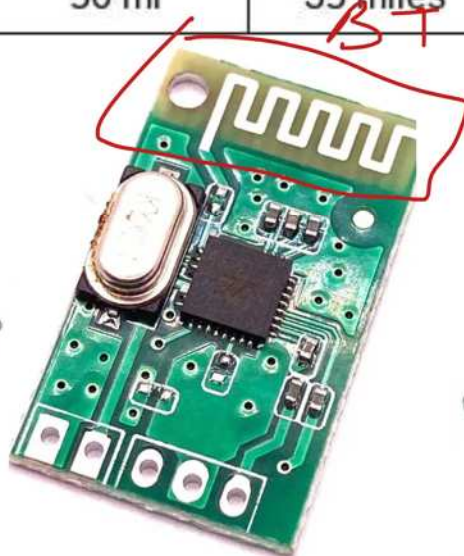
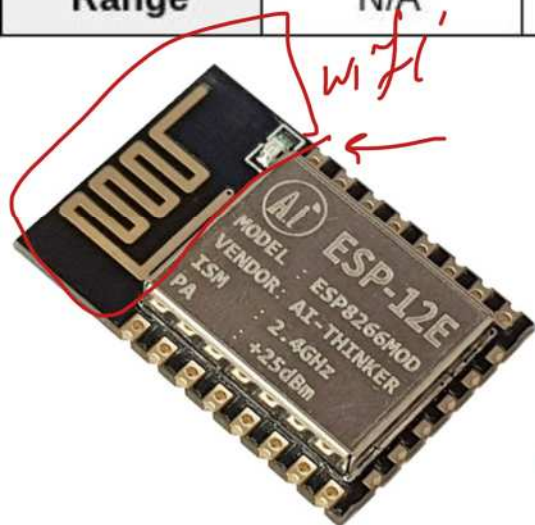
115 kHz

برای سیگنال

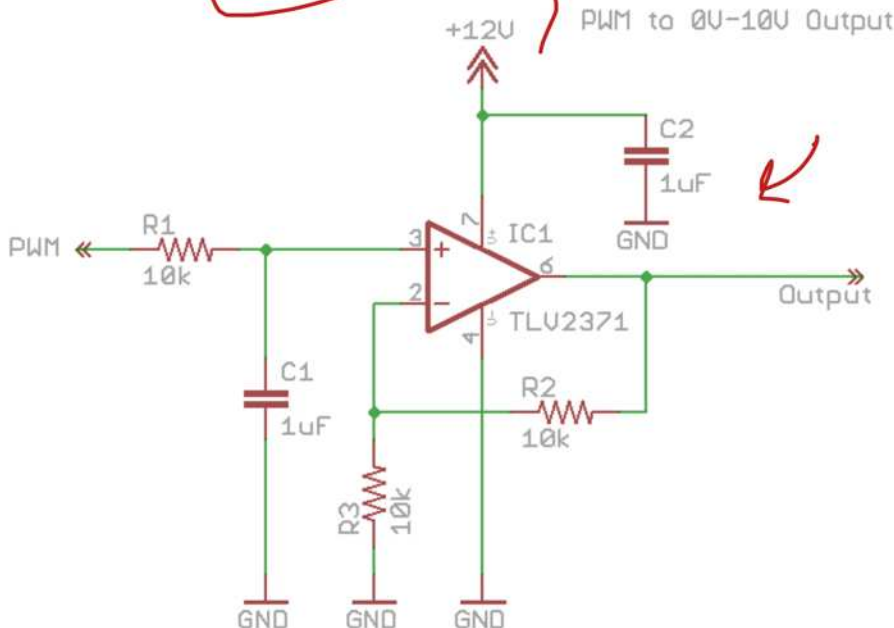
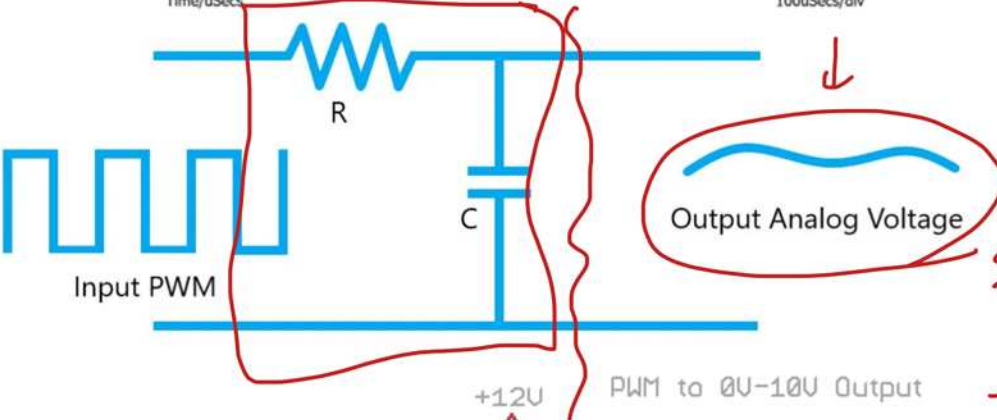
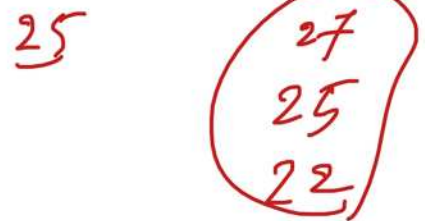
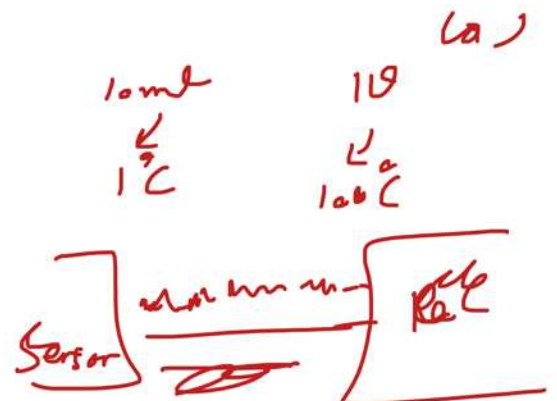
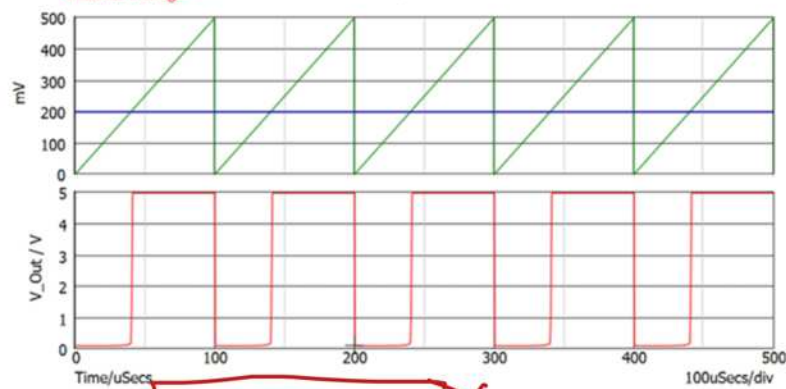
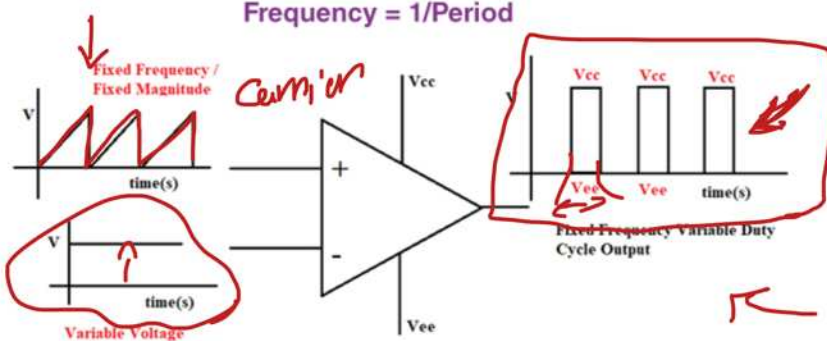
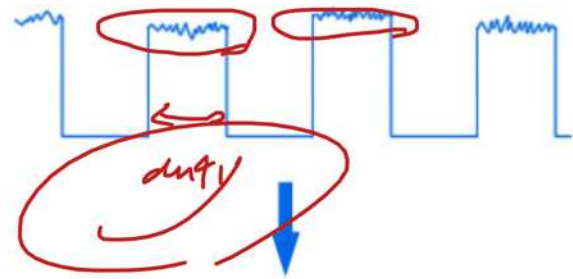
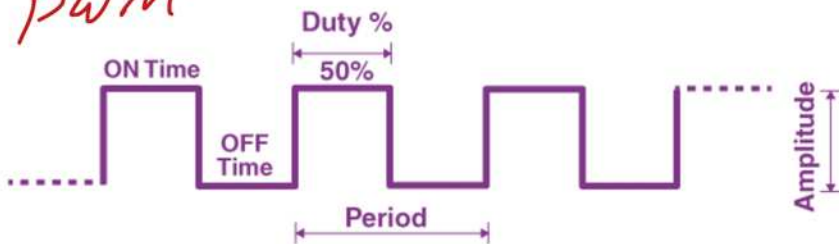


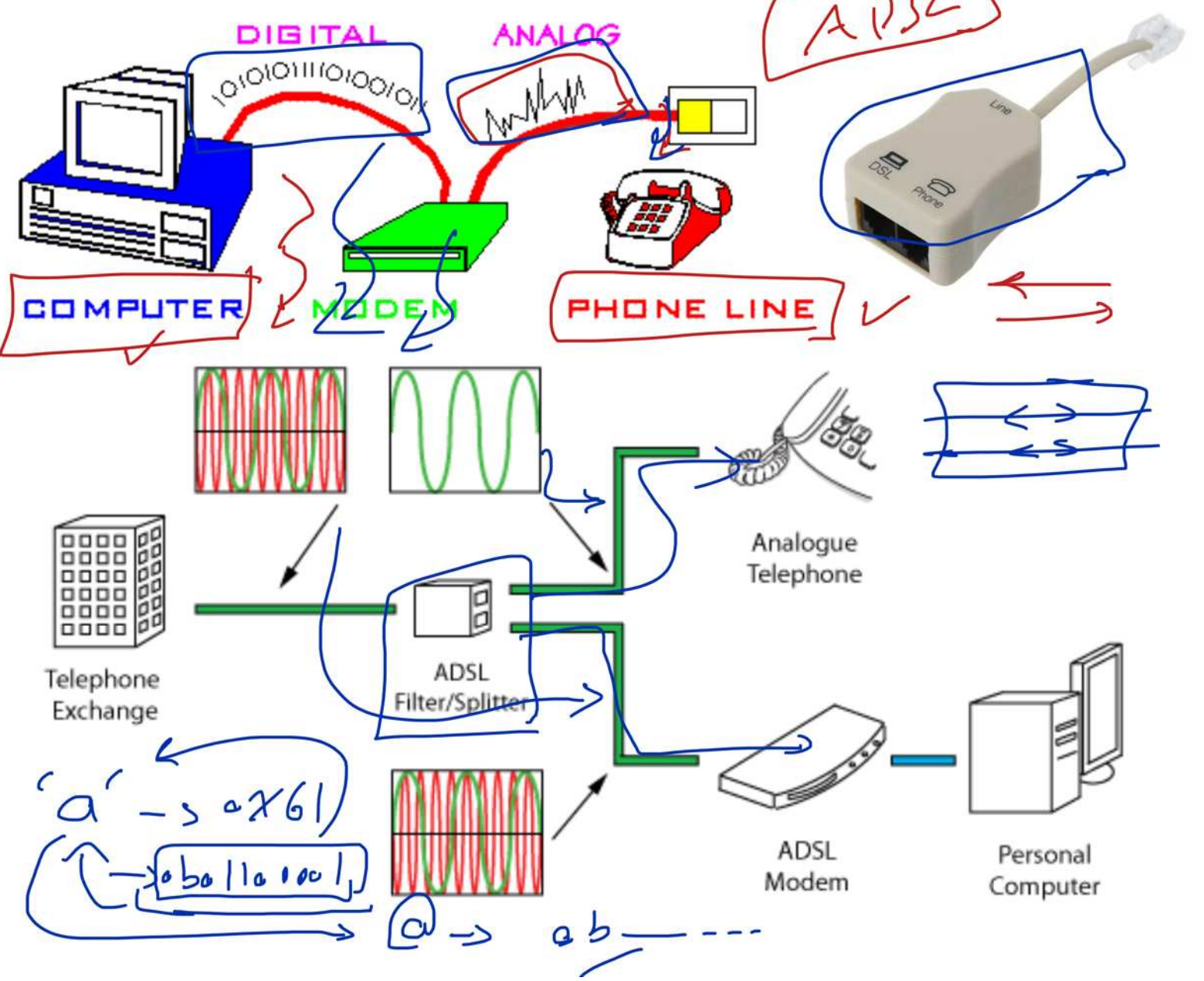
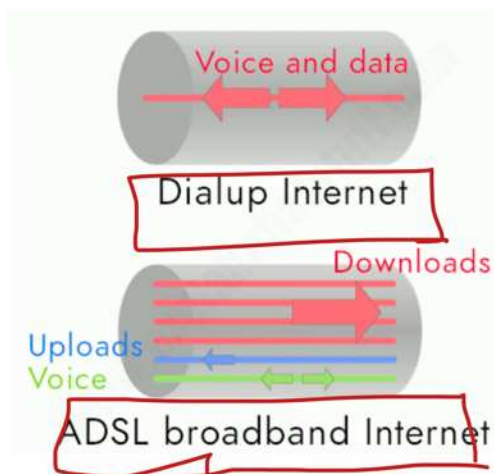
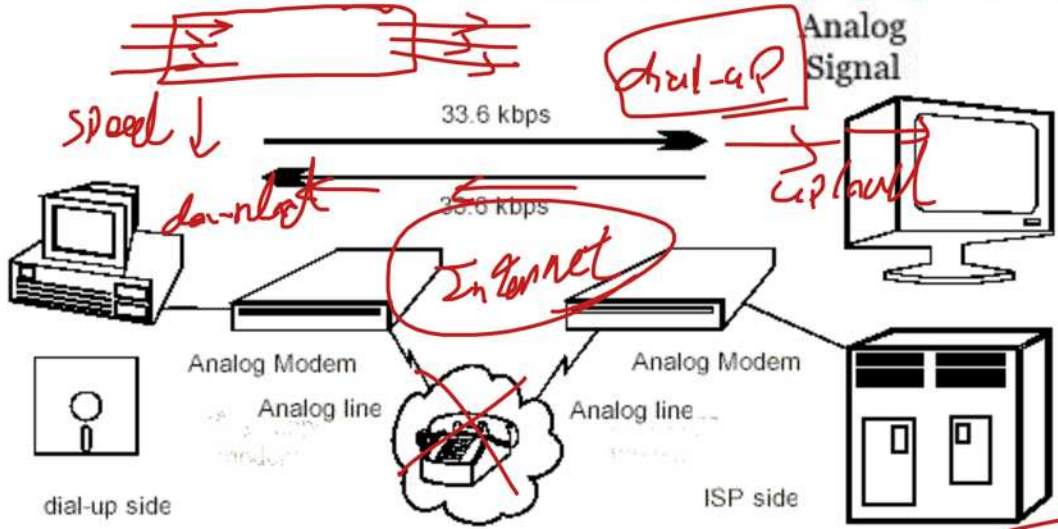
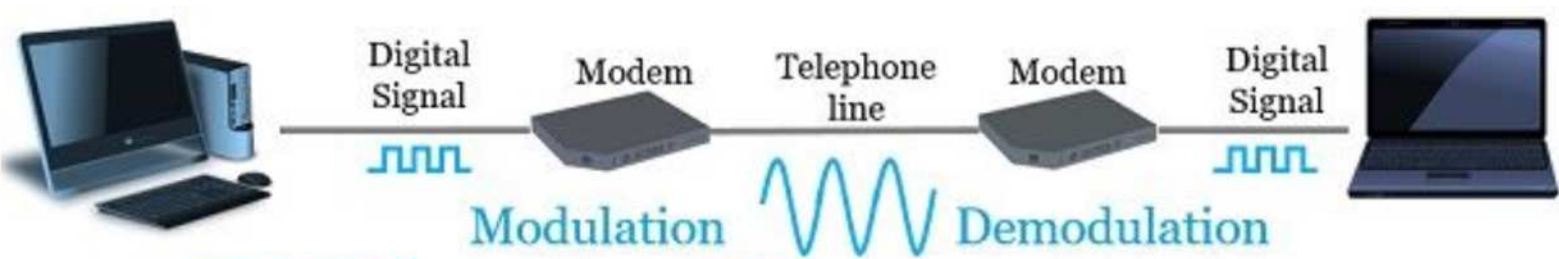


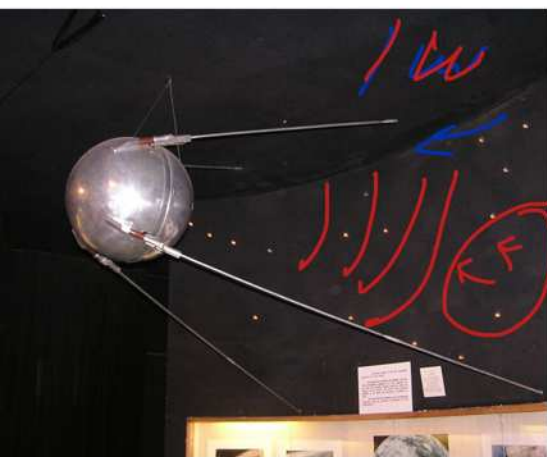
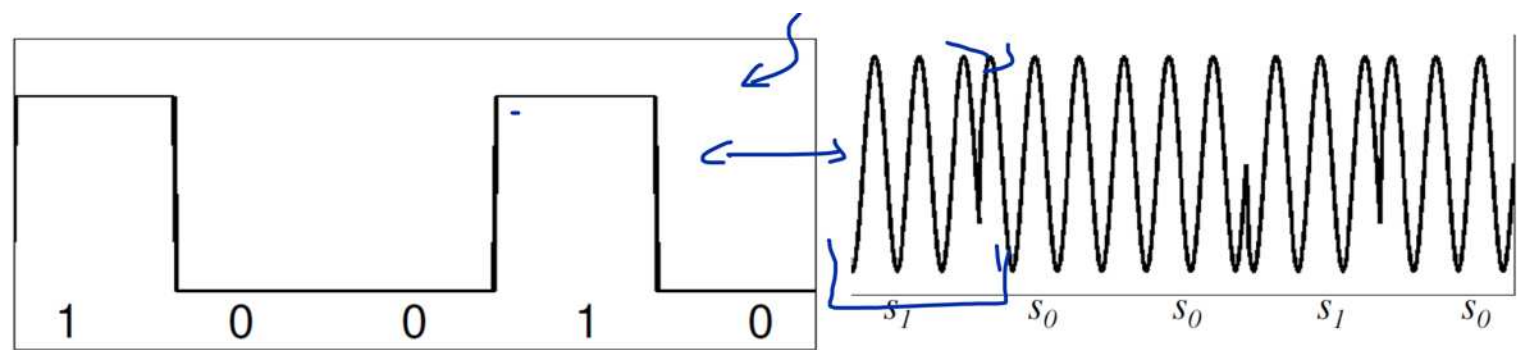
Technology	1G	2G	3G	4G	5G
Year	1979	1991	2001	2009	2019
Use Cases	Analog System, Dropped Calls, Giant Cell Phones	Texting (SMS), MMS, Conference Calls, Long Distance Call Tracking	Cheap data transmission, GPS, Web Browsing, SD Video Streaming	HD Video Streaming, Wearable Devices, High Speed Applications	Internet of Things, Cloud Storage, Remote Surgical Robots
Frequency	30 KHz	1.8 GHz	1.6-2 GHz	2-8 GHz	3-30 GHz
Bandwidth	2 kbps	364 kbps	3 Mbps	100 Mbps	10 Gbps
Avg Speeds	2 kbps	40 kbps	300 kbps	25 Mbps	150 Mbps
Range	N/A	50 mi	35 miles	10 miles	1,000 ft



pwm

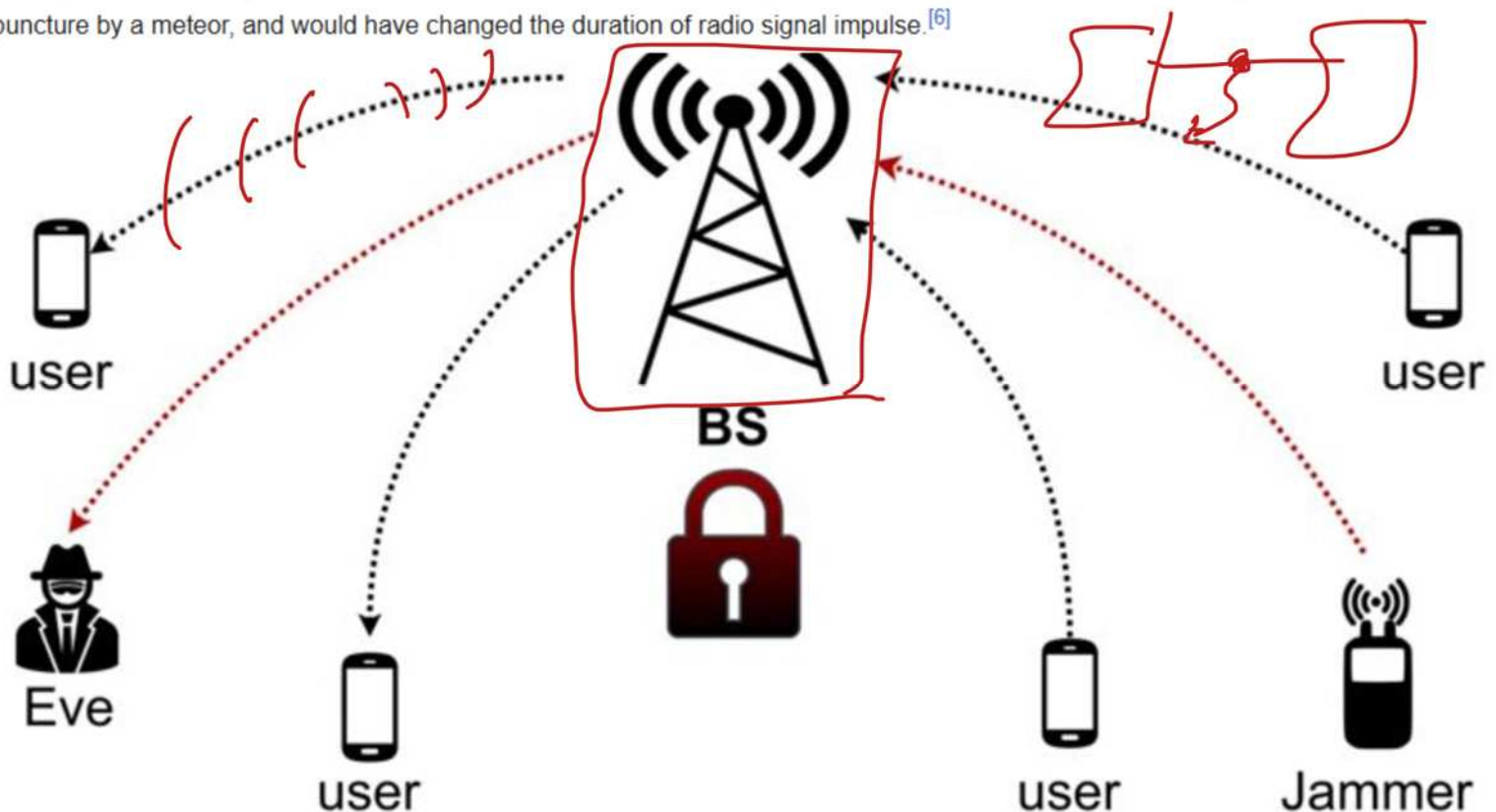


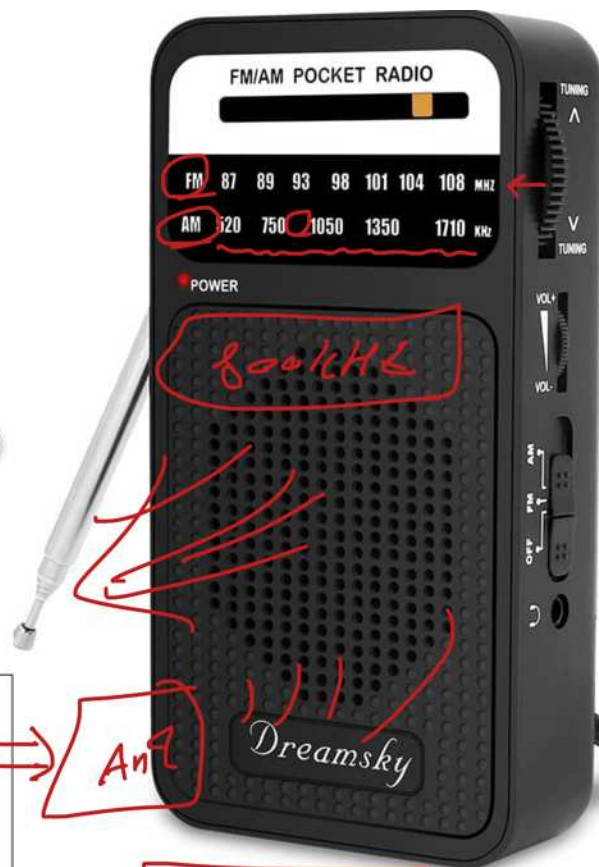
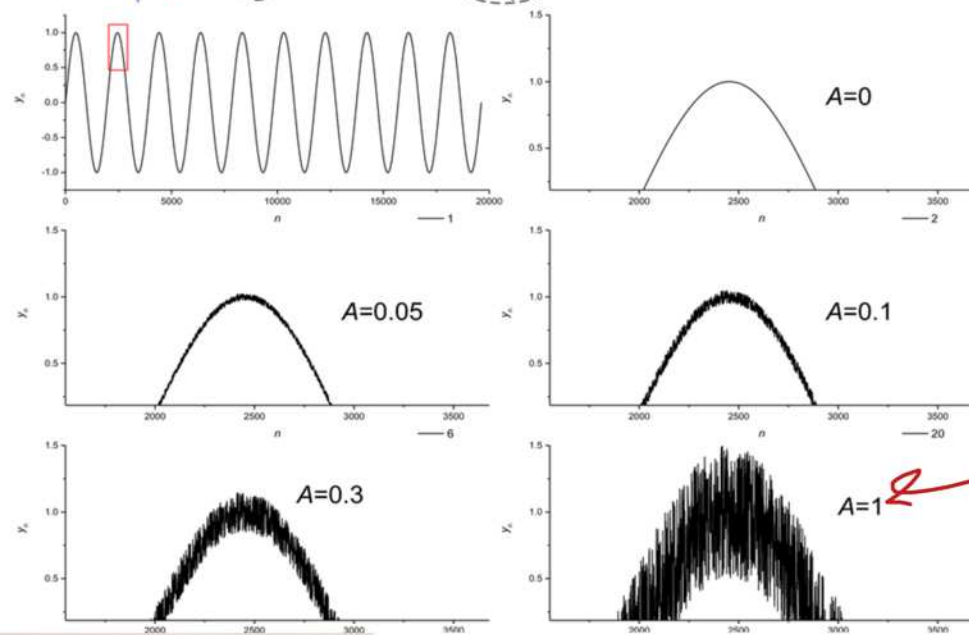
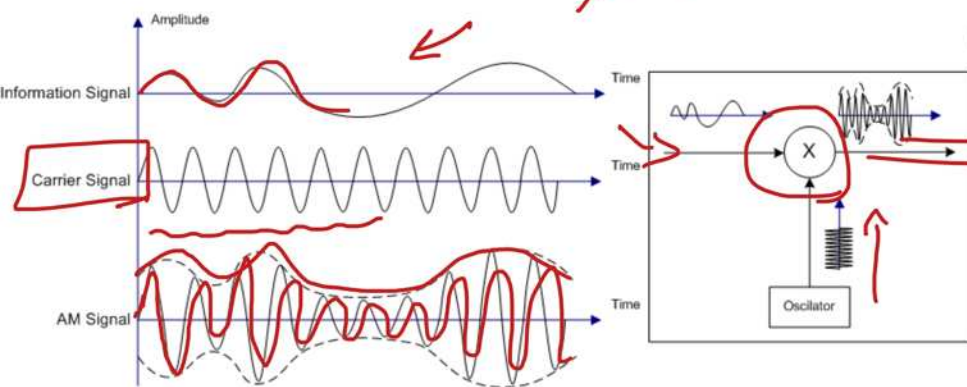
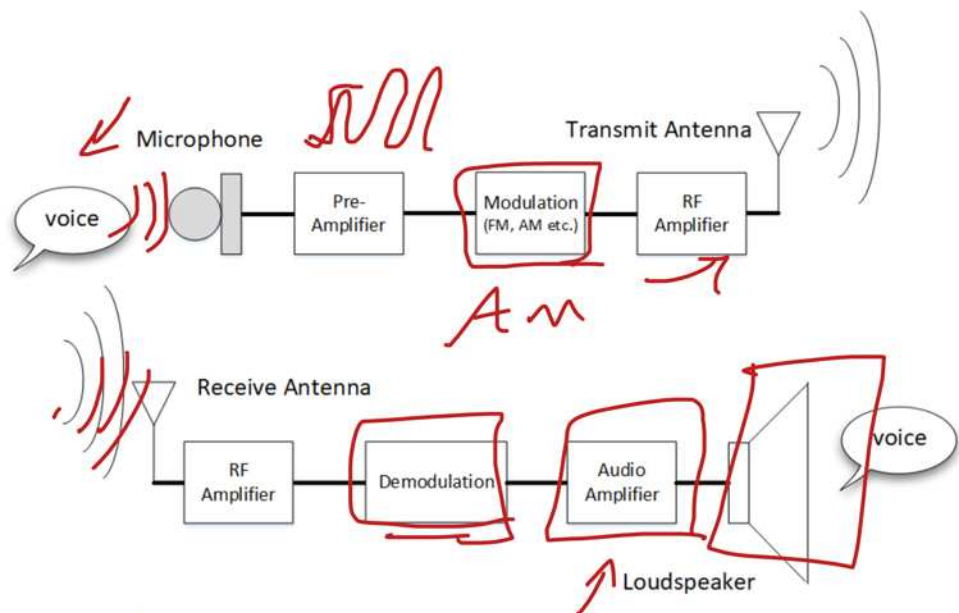




Sputnik 1 was launched during the [International Geophysical Year](#) from [Site No. 1/5](#), at the 5th [Tyuratam](#) range, in [Kazakh SSR](#) (now known as the [Baikonur Cosmodrome](#)). The satellite travelled at a peak speed of about 8 km/s (18,000 mph), taking 96.20 minutes to complete each orbit. It transmitted on 20.005 and 40.002 MHz,^[11] which were monitored by radio operators throughout the world. The signals continued for 22 days until the transmitter batteries depleted on 26 October 1957. On 4 January 1958, after three months in orbit, Sputnik 1 burned up while [reentering Earth's atmosphere](#), having completed 1,440 orbits of the Earth,^[3] and travelling a distance of approximately 70,000,000 km (43,000,000 mi).^[12]

The satellite had a one-watt, 3.5 kg (7.7 lb)^[38] radio transmitting unit inside, developed by Vyacheslav I. Lappo from *NII-885*, the Moscow Electronics Research Institute,^{[57][58]} that worked on two frequencies [20.005 and 40.002 MHz](#). Signals on the first frequency were transmitted in 0.3 s pulses (near $f = 3$ Hz) (under normal temperature and pressure conditions on board), with pauses of the same duration filled by pulses on the second frequency.^[59] Analysis of the radio signals was used to gather information about the electron density of the ionosphere. Temperature and pressure were encoded in the duration of radio beeps. A temperature regulation system contained a [fan](#), a dual [thermal switch](#), and a control thermal switch.^[57] If the temperature inside the satellite exceeded 36 °C (97 °F), the fan was turned on; when it fell below 20 °C (68 °F), the fan was turned off by the dual thermal switch.^[55] If the temperature exceeded 50 °C (122 °F) or fell below 0 °C (32 °F), another control thermal switch was activated, changing the duration of the radio signal pulses.^[57] Sputnik 1 was filled with dry nitrogen, pressurised to 1.3 atm (130 kPa).^[40] The satellite had a [barometric switch](#), activated if the pressure inside the satellite fell below 130 kPa, which would have indicated failure of the pressure vessel or puncture by a meteor, and would have changed the duration of radio signal impulse.^[6]

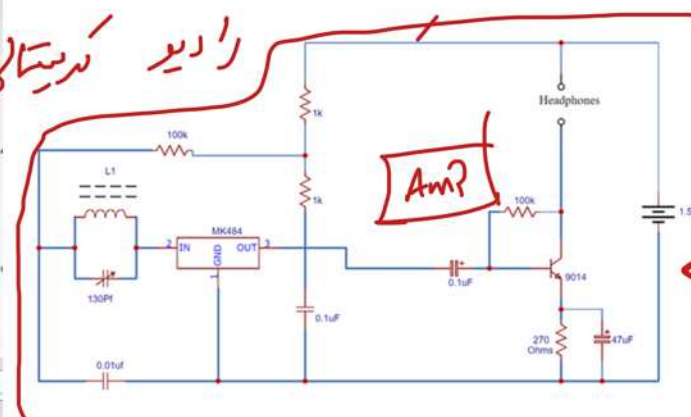




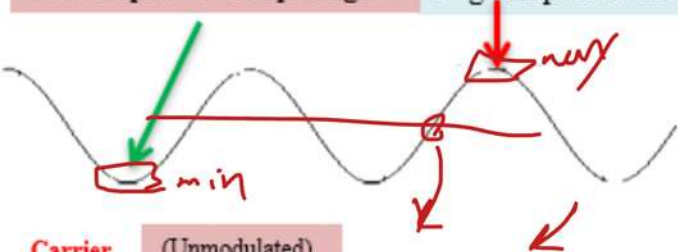
Am



راديو دستی

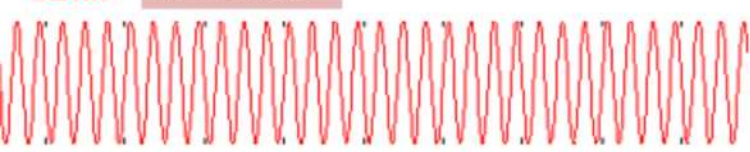


High amplitude of input signal



FM

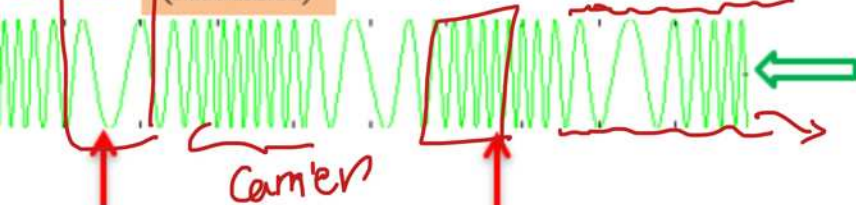
Carrier (Unmodulated)



Unmodulated carrier has constant amplitude, constant frequency and constant phase.

Fact

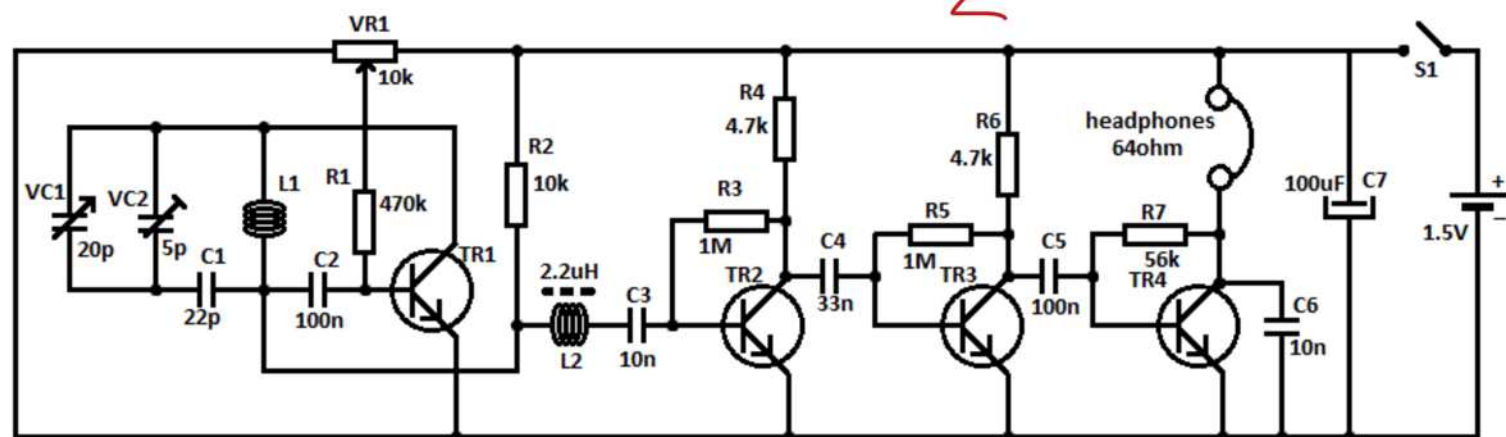
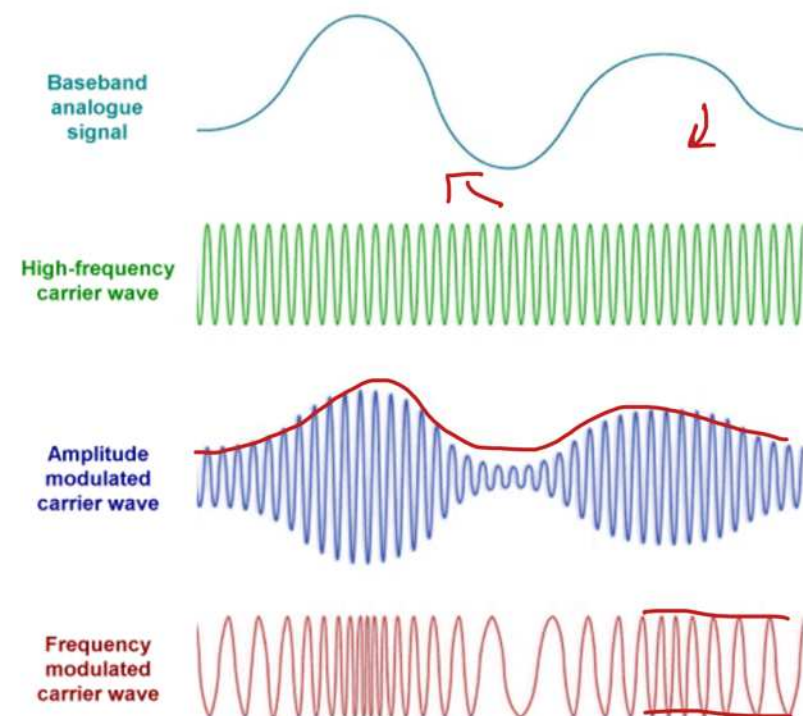
FM wave (Modulated)



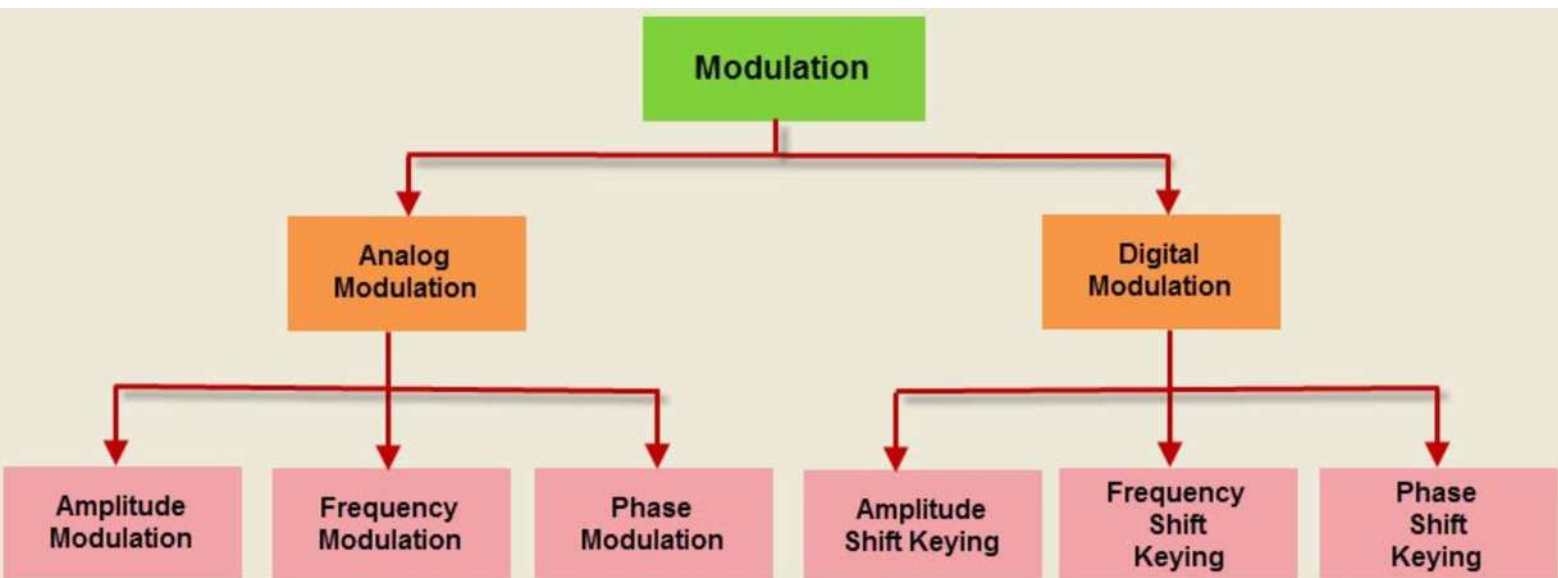
FM signal has constant amplitude and constant phase, but frequency changes according to **amplitude of input signal**.

Low frequency change

High frequency change

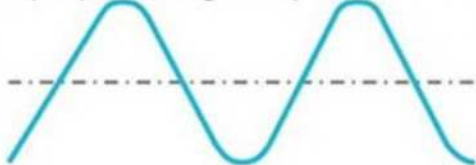


FM Radio Receiver

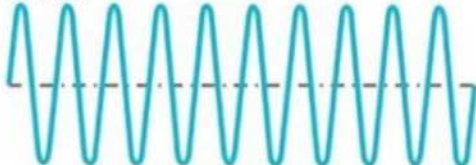


Amplitude Modulation (AM)

Input (Modulating Wave)



Carrier

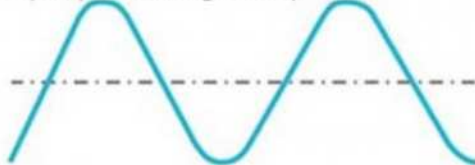


Modulated Result

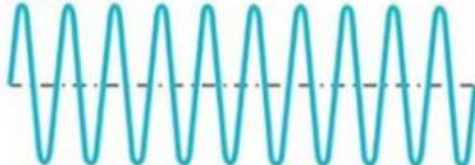


Frequency Modulation (FM)

Input (Modulating Wave)



Carrier

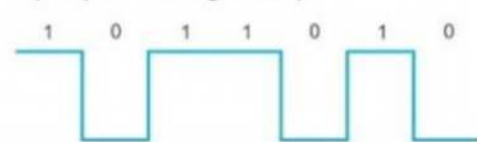


Modulated Result



Digital Modulation

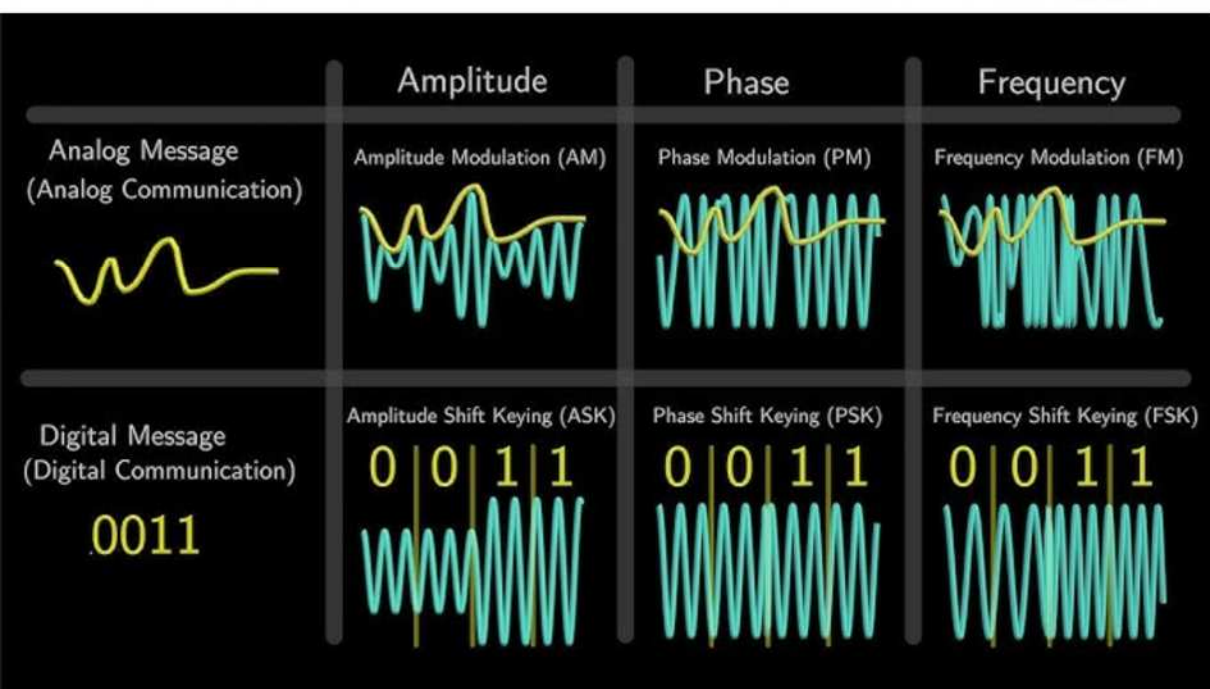
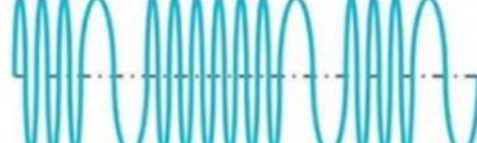
Input (Modulating Wave)

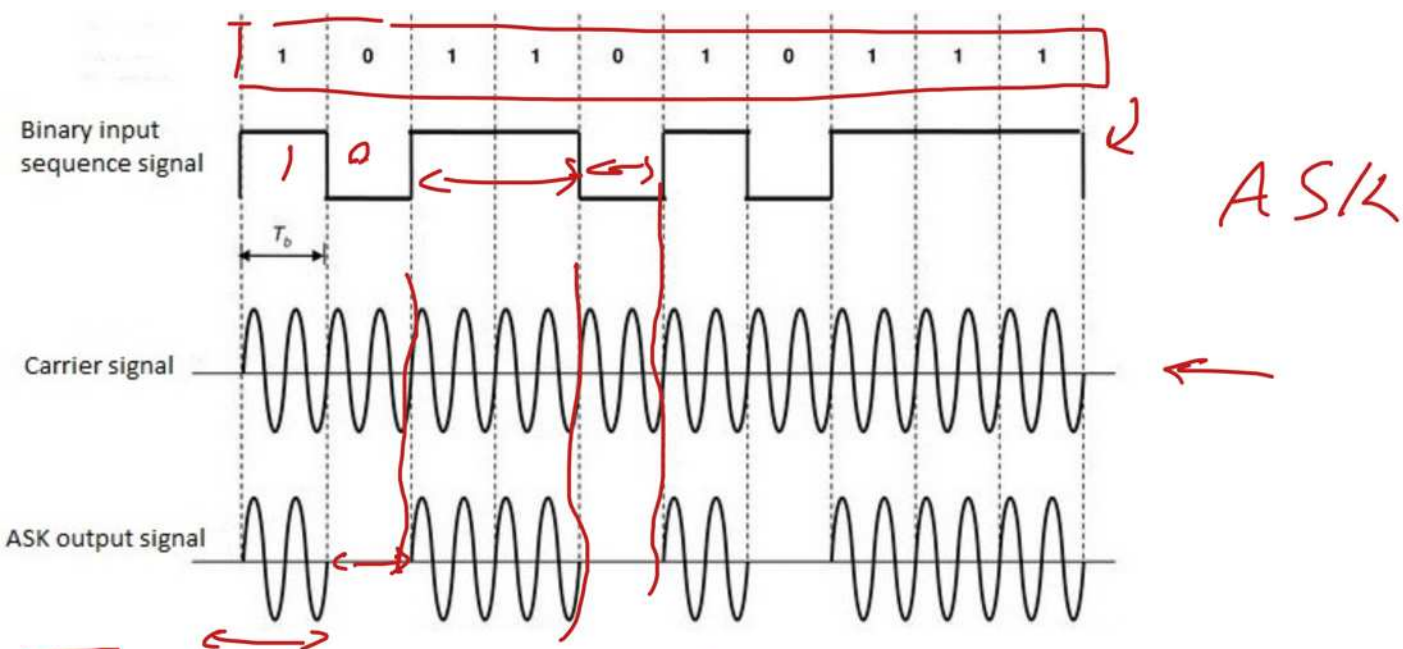


Carrier



Modulated Result

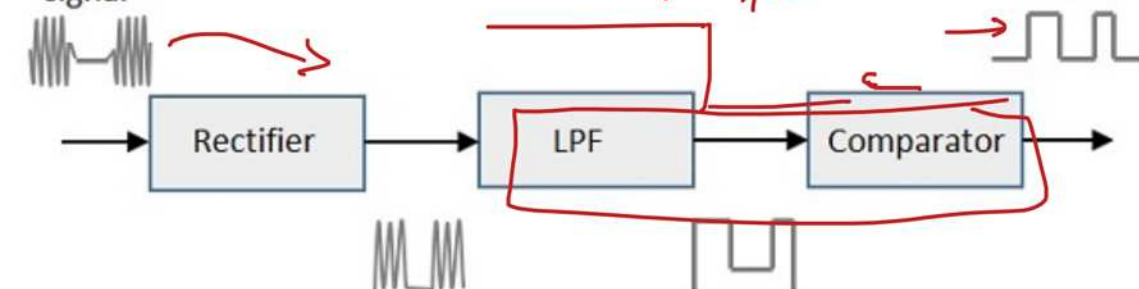
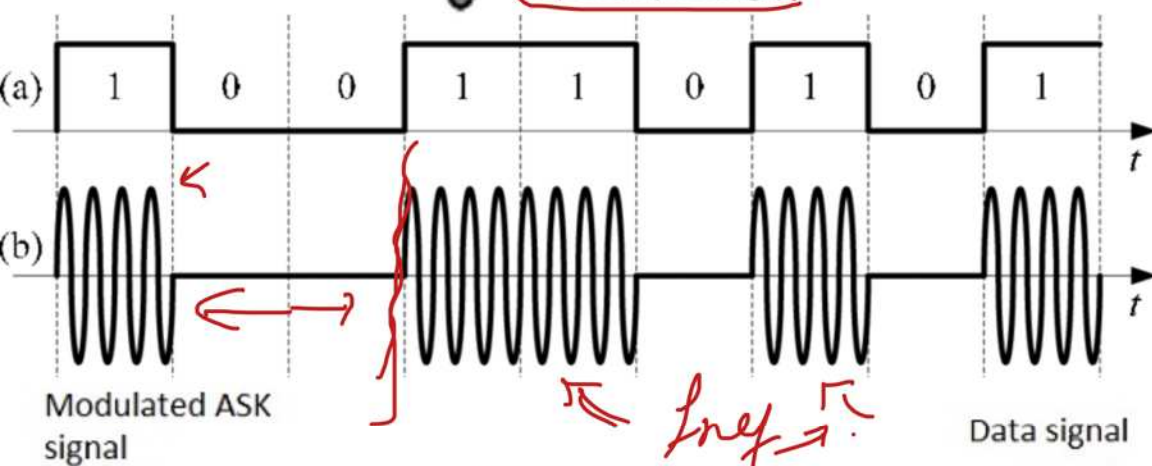
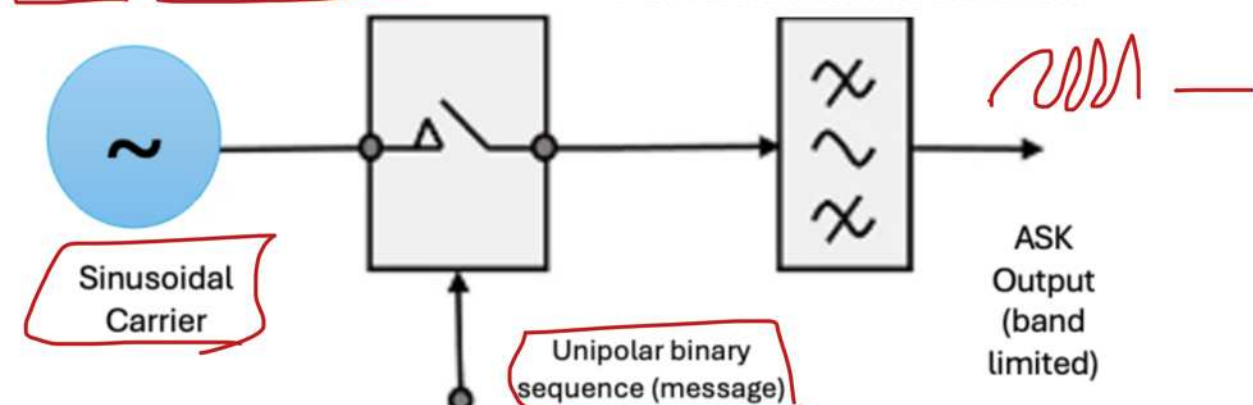




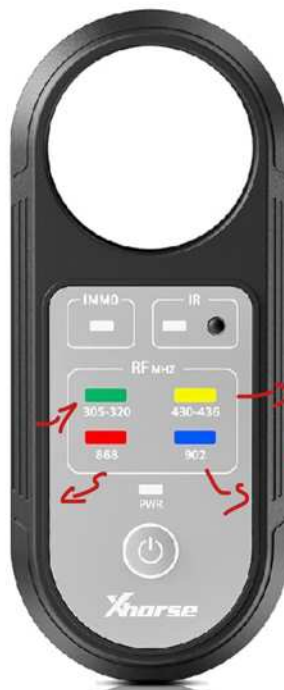
ASK: Amplitude Shift Keying

OOK: On-off keying

ASK Generation method



Asynchronous ASK detector

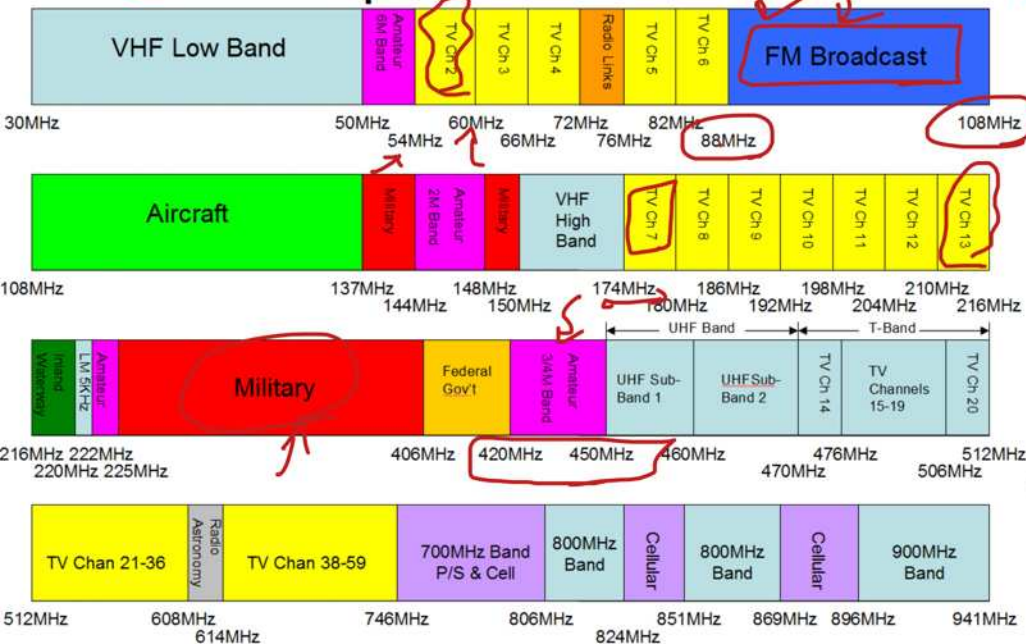


315 MHz

433 MHz

Testo!

Spectrum View



Installation

Transponder

28	4000	H	27500
29	3908	H	27500
30	4000	V	27500
31	3908	V	27500
32	4000	H	30000

S

61%

Q

95%

Add

Edit

Delete

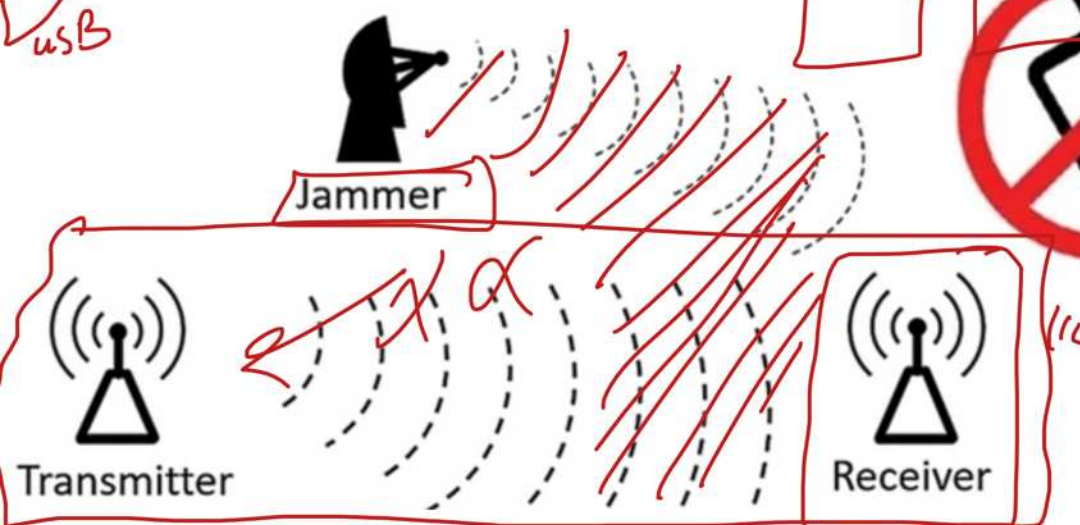
Scan

SAT Satellite

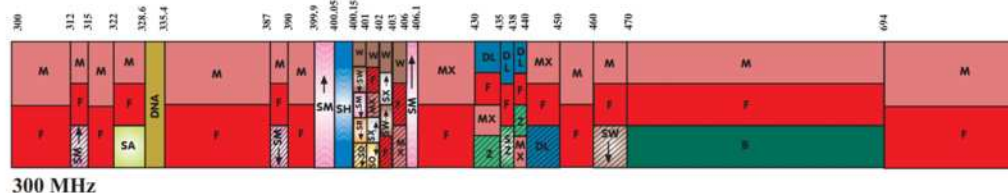
OK Select

INFO Info

EXIT Exit



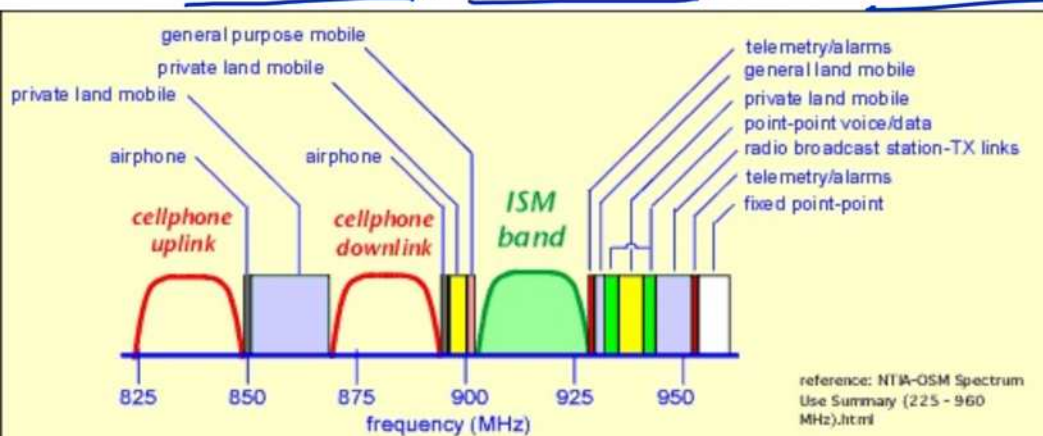
جدول تخصیص فرکانس امواج رادیویی جمهوری اسلامی ایران (باند UHF)



300 MHz



ISM: Industrial, Scientific, and Medical



سرویسهای رادیویی

سرویسهای رادیویی	Radiocommunication Services
متحرک ماهواره‌ای هوانوردی (R)	SMAR Aeronautical Mobile-satellite (R)
متحرک دریایی ماهواره‌ای	SMM Maritime mobile satellite
پخش همگانی	B Broadcasting
تعیین موقعیت رادیویی	DL Radiolocation
ناوبری رادیویی	DN Radionavigation
ناوبری رادیویی هوانوردی	DNA Aeronautical radionavigation
ناوبری رادیویی دریایی	DNM Maritime radionavigation
ثابت	F Fixed
فرکانس استاندارد و سیگنال زمان	W Standard frequency and time signal
متحرک	M Mobile
متحرک خشکی	MT Land mobile
متحرک دریایی	MM Maritime mobile
متحرک هوانوردی	MA Aeronautical mobile
متحرک هوانوردی (R)	MAR Aeronautical mobile (R)
متحرک هوانوردی (OR)	MAO Aeronautical mobile (OR)
متحرک بجز متحرک هوانوردی	MX Mobile except aeronautical mobile
متحرک بجز متحرک هوانوردی (R)	MXR Mobile except aeronautical mobile (R)
کمک هواشناسی	W Meteorological aids
آماور	Z Amateur
کیهان شناسی رادیویی	SA Radio astronomy
پخش همگانی ماهواره‌ای	B Broadcasting-satellite
تعیین رادیویی ماهواره‌ای	SD Radiodetermination-satellite
ناوبری رادیویی ماهواره‌ای	SDN Radionavigation-satellite
ثابت ماهواره‌ای	SF Fixed-satellite
فرکانس استاندارد و سیگنال زمان ماهواره‌ای	SW Standard frequency and time signal-satellite
بین ماهواره‌ای	IS Inter-satellite
متحرک ماهواره‌ای	SM Mobile-satellite
عملیات فضایی	SO Space operation
تحقیق فضایی	SR Space research
هواشناسی ماهواره‌ای	SW Meteorological-satellite
اکتشاف زمین ماهواره‌ای	SX Earth exploration-satellite
آماور ماهواره‌ای	SZ Amateur-satellite
متحرک ماهواره‌ای بجز متحرک دریایی ماهواره‌ای	SMY Mobile-satellite except maritime mobile satellite